SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO1: Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1,BL2
Assessment Tool Weightage: 30%

QUESTIONS: 2 **Total Marks:50**

Annexure A

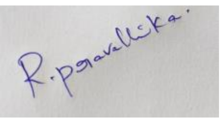
Technical Presentation

Criteria	***Technical Content Accuracy(3)***	***Problem Identification(2.5)***	***Use of Tools(2)***	***Communication(1.5)***	***Professionalism(1)***	***Total(10)***
Weightage	*30%*	25%	20%	20%	10%	*100%*
Q1						
Q2						

Code of Conduct:

*I **RAJANEDI DATI PRAVALLIKA(192525473**) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Technical Presentation					
Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Annexure A

Topic 1: Decision Support Systems (DSS)

▯ Focus Areas:

- Definition and need of DSS
- Components (Data, Model, UI)
- Types of DSS
- Real-world applications (banking, healthcare)
- Advantages & limitations

Topic 2: Operational Systems vs DSS

▯ Focus Areas:

1. OLTP vs OLAP concepts
2. Key differences (data, users, purpose)
3. Performance comparison
4. Use-case scenarios
5. Integration between systems

Topic 3: Architecture of Data Warehouse

▯ Focus Areas:

1. Data warehouse definition
2. Three-tier architecture
3. ETL process
4. Data marts & metadata
5. Benefits in decision-making

Topic 4: OLAP and Data Cubes

⍰ Focus Areas:

1. OLAP concept
2. Data cube structure
3. OLAP operations (slice, dice, roll-up, drill-down)
4. Applications in analytics

Topic 5: Dimensional Data Modeling

⍰ Focus Areas:

1. Fact and dimension tables
2. Star schema vs snowflake schema
3. Schema design process
4. Real-world example

Topic 6: Query Processing & BI Tools

⍰ Focus Areas:

1. Query processing stages
2. Query optimization
3. BI tools overview
4. Reporting and visualization
5. Examples of open-source BI tools

Topic 7: OLAP Application using KNIME

⍰ Focus Areas:

1. Introduction to KNIME
2. Workflow components (nodes, connections)
3. Performing OLAP operations
4. Sample application/demo

5. Report generation

PART B: PRESENTATION RULES & GUIDELINES

? 1. Group Formation

1. Each group: 3–5 students
2. Each member must participate
3. Assign roles:
    1. Presenter
    2. Designer (PPT)
    3. Researcher
    4. Demo/Case handler

? 2. Presentation Structure (MANDATORY)

Each group must include:

1. Introduction
    1. Definition and importance
2. Core Concepts
    1. Explanation with diagrams
3. Process/Architecture
    1. Step-by-step explanation
4. Real-World Example
    1. Case study or application
5. Tool/Demo (if applicable)
    1. KNIME or BI tool
6. Conclusion
    1. Summary + future scope

? 3. Slide Preparation Rules

1. Maximum slides: 10–15
2. Use:
    1. Diagrams
    2. Flowcharts

    3. Tables (for comparison)
3. Avoid:
    1. Too much text
    2. Copy-paste content

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4. Presentation Delivery Rules

1. Time limit: 10–12 minutes per group
2. Each member must speak at least 2 minutes
3. Maintain:
    1. Eye contact
    2. Clear voice
    3. Logical flow

Business Intelligence Using KNIME and OLAP

Course Name: CSA1607 SLOT D-DWDM

Guided by :
Dr.D BEULAHDAVID

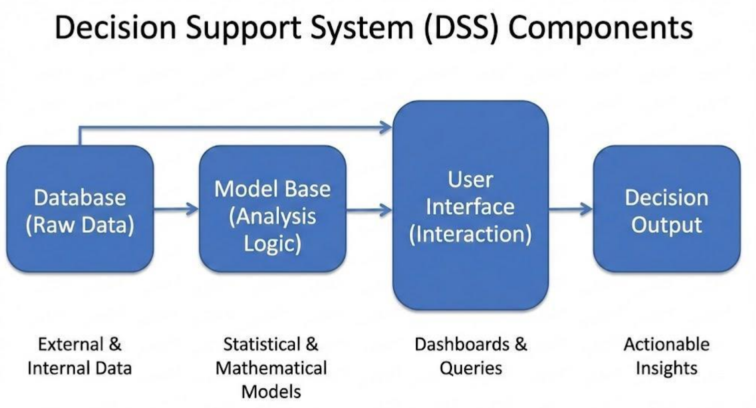
Presented By:
R.DATI PRAVALLIKA(192525473)

Introduction

- Business Intelligence (BI) converts raw data into useful information.
- Helps organizations make better decisions.
- Improves productivity and business performance.
- Supports data-driven decision making.
- Uses Data Warehouses, ETL, OLAP, and BI tools like KNIME

Decision Support System (DSS)

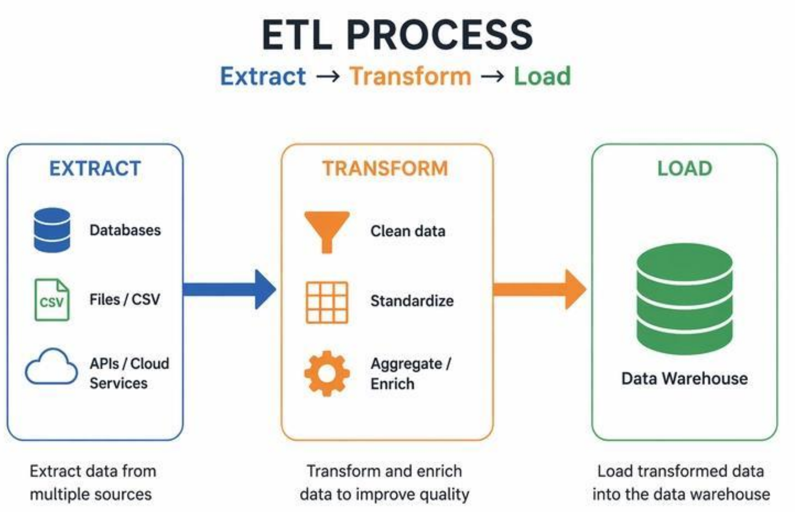
- Helps managers make informed decisions.
- Analyzes historical and current data.
- Supports planning and forecasting.
- Solves business problems using data analysis.
- Example: Sales forecasting.



Data Warehouse & ETL

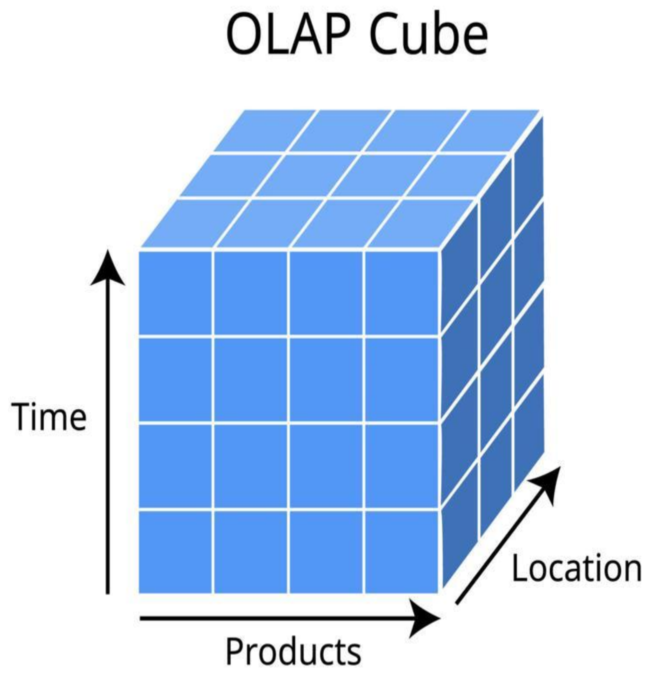
- Data Warehouse stores integrated business data.
- Collects data from multiple sources.
- Supports reporting and analytics.
- **ETL Process:**

  Extract – Collect data.
  Transform – Clean and organize data.
  Load – Store data in the warehouse.



OLAP & Data Cubes

- OLAP stands for Online Analytical Processing.
- Performs multidimensional data analysis.
- Helps identify trends and patterns
- **OLAP Operations:**
  Slice
  Dice
  Drill Down
  Roll Up

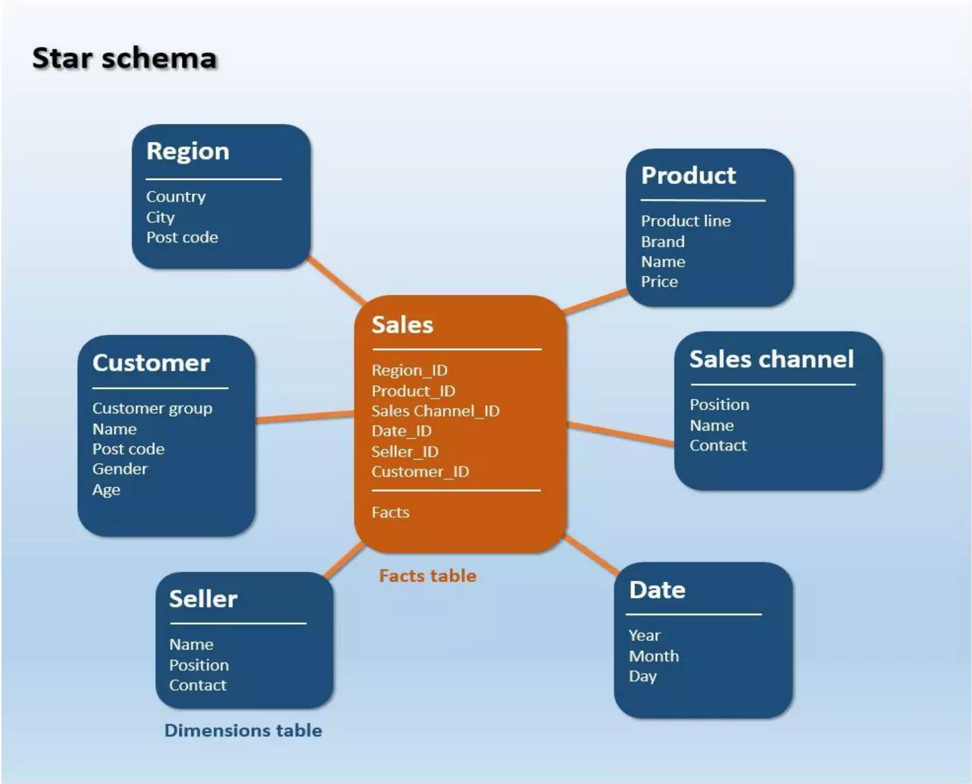


Star Schema, Snowflake Schema & KNIME

- **Star Schema:**
  One fact table.
  Multiple dimension tables.

- **Snowflake Schema:**
  Normalized dimension tables.
  Reduces redundancy.

- **KNIME:**
  Open-source analytics tool.
  Drag-and-drop workflow.

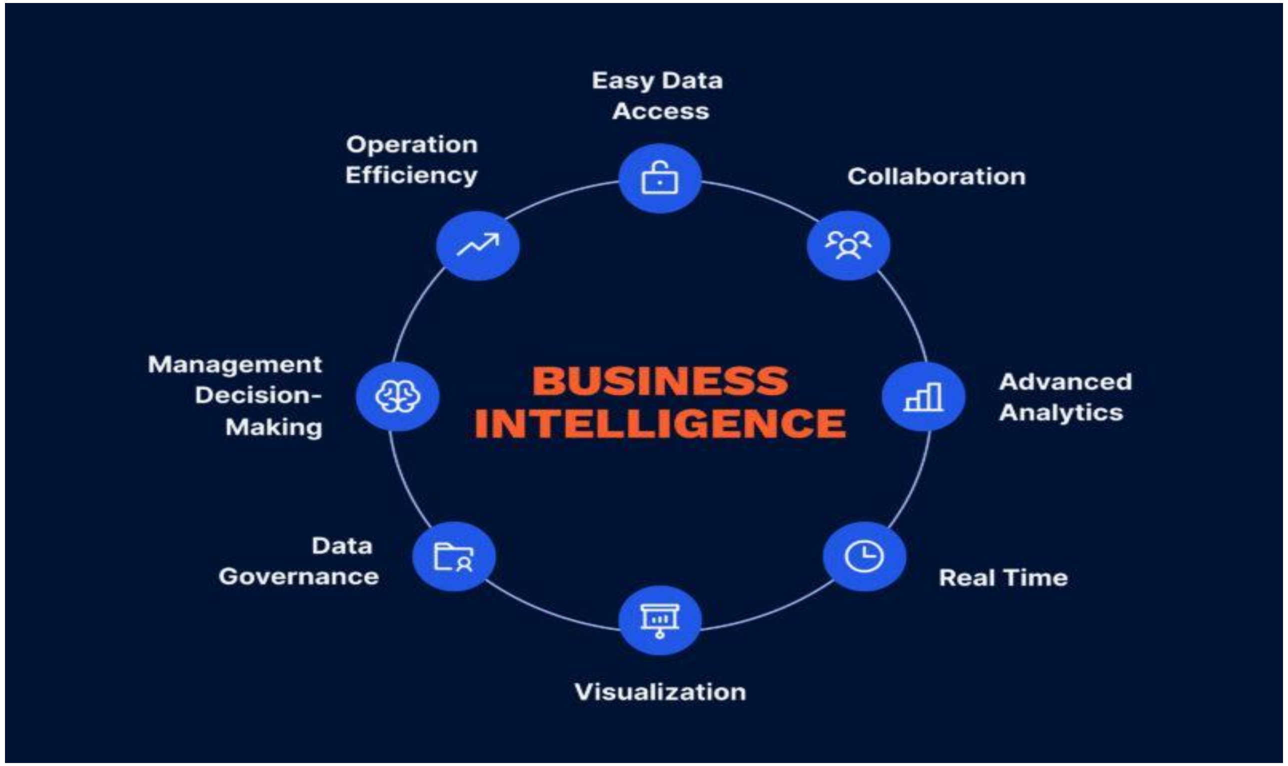


Applications, Advantages & Conclusion

Engineer to Excel

- **Applications:**

Banking
Healthcare
Retail



- **Advantages:**

Better decision making.
Faster data analysis.

- **Conclusion:**

Business Intelligence transforms data into insights.
ETL, Data Warehousing, and OLAP improve analytics.